

Notes: Chen, Evans & Sun (JFE, 2025)
Self-Declared Benchmarks and Fund Manager Intent: “Cheating” or Competing?

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1 Big picture

- **Question.** Mutual funds choose self-declared prospectus benchmarks that investors use for relative performance evaluation. Are mismatched benchmarks mainly strategic manipulation—“cheating”—or can they also reflect legitimate product-positioning and benchmarking difficulties?
- **Setting.** The paper studies U.S. active equity mutual funds using Morningstar benchmark snapshots from December 2008 to December 2020. The panel contains 3,709 unique actively managed equity funds, 154 individual benchmarks, and 806 primary benchmark changes.
- **Core fact.** Benchmark mismatch is high in 2008, but it declines meaningfully over time: mismatched assets fall from about 45.4% in 2008 to about 27.8% in 2020, and mismatched funds fall from 34.2% to 27.2%.
- **Main mechanism.** The decline is mostly intensive-margin correction by existing funds, not merely the entry of matched funds or exit of mismatched funds. Among funds mismatched in 2008 and surviving to 2020, roughly 58% correct their benchmark.
- **New variable.** The paper introduces **Investment Strategy Breadth**, or **IS Breadth**: the percentage of a fund’s holdings that lie outside its Morningstar style-box category. High IS Breadth means the fund is harder to summarize with one standard benchmark.
- **Key distinction.** For specialized funds, mismatches look more like performance gaming: mismatched benchmarks have lower factor loadings or lower subsequent returns than the matched benchmarks funds later adopt. For broad funds, mismatches look less like gaming and more like a risk-hedging or product-positioning choice.
- **Investor-learning channel.** As investors become better able to identify incorrect benchmarks, specialized mismatched funds are more likely to switch to a matched benchmark.
- **Governance channel.** Institutional versions or separate-account twins appear to improve benchmark quality, especially for specialized funds, because institutional clients are better monitors.
- **Competition channel.** More competition within a benchmark category, measured by the number or market share of index funds, increases the probability that funds switch benchmarks and switch to more matched benchmarks.
- **Product-characteristics channel.** Funds switch benchmarks when the new benchmark improves relative performance or reduces tracking error; this suggests that benchmark choice is shaped both by return presentation and by risk positioning.
- **Managerial-entrenchment result.** Longer manager tenure and greater family-level assets under management by the manager are associated with greater persistence of mismatched benchmarks, especially for specialized funds.
- **Takeaway.** Not every mismatch is cheating. Benchmark accuracy improves through market forces—investor learning, institutional oversight, and competition—but those forces are strongest when the investment strategy is narrow enough that the “right” benchmark is relatively clear.

2 Data and measurement

2.1 Benchmark data and sample construction

- The paper uses Morningstar database snapshots collected between 2008 and 2020. Each snapshot lists a fund’s self-designated prospectus benchmark and benchmark characteristics at that time.
- The sample is active U.S. equity mutual funds. The authors rely on Morningstar equity style classifications for both funds and benchmarks.

- The final monthly panel includes fund characteristics, holdings-implied style categories, benchmark indicators, fund returns, flows, Morningstar ratings, family assets, expenses, loads, and age.
- The authors focus on the *primary* benchmark because it is the most visible benchmark in Morningstar and investor materials and is the one most likely to influence investor interpretation.

Interpretation: The data are unusually well suited for this question because they observe benchmarks dynamically, not only in a single cross-section. This allows the paper to distinguish persistent mismatching from active benchmark correction.

2.2 Benchmark mismatch

The main mismatch indicator follows the logic of Sensoy (2009). A fund is coded as mismatched if its self-designated benchmark has a Morningstar style classification different from the style implied by the fund's holdings:

$$\text{BenchmarkMismatch}_{i,t} = 1 \quad \text{if benchmark style}_{i,t} \neq \text{holdings-implied style}_{i,t}.$$

- Example: a fund whose holdings look like Large Cap Value but whose benchmark is the S&P 500 is classified as mismatched.
- The classification uses size categories such as Large, Mid, and Small and growth-value categories such as Growth, Blend, and Value.
- The measure is intentionally simple and transparent, but the paper recognizes that some funds have broad strategies that do not fit cleanly into one style box.

Interpretation: The mismatch indicator is powerful because it measures a concrete difference between the fund's benchmark and its actual holdings. But it can overstate opportunism for funds whose strategy is deliberately broad.

2.3 Investment Strategy Breadth

The paper constructs IS Breadth using fund holdings and Morningstar style boxes:

$$\text{ISBreadth}_{i,t} = \frac{\text{equity holdings outside the fund's core style box}}{\text{total equity holdings}}.$$

The standardized version is

$$\text{StdISBreadth}_{i,t} = \frac{\text{ISBreadth}_{i,t} - \overline{\text{ISBreadth}}}{\sigma(\text{ISBreadth})}.$$

- IS Breadth captures how difficult it is to summarize a fund with a single style benchmark.
- Specialized funds have low IS Breadth. Broad funds have high IS Breadth.
- The authors validate the measure using fund names, style drift, and mandate descriptions.

Interpretation: This variable is central because it separates two economically distinct cases: a specialized fund picking a misleading benchmark versus a broad fund choosing among imperfect benchmarks.

2.4 Bias and variance of benchmark mismatch

For mismatched funds, the paper compares the self-declared benchmark with the best-fitting benchmark among popular equity benchmarks.

- **Bias** is the average difference between the fund’s return relative to the best-fitting benchmark and the fund’s return relative to its self-declared benchmark.
- **Variance** is the squared difference between the monthly returns of the self-declared benchmark and the best-fitting benchmark over a 36-month window.

Interpretation: Positive bias suggests the chosen benchmark makes the fund look better. High variance suggests the chosen benchmark may be used to hedge different risk exposure or to represent a broad investment mandate.

2.5 Flow and performance variables

- Investor flows are measured as percent flows, positive-flow indicators, and changes in market share.
- Benchmark-adjusted return is the fund return minus the self-declared benchmark return.
- Fund alpha is the 36-month Fama–French three-factor alpha using rolling factor estimates.
- Flow beta measures the sensitivity of fund flows to benchmark-adjusted returns.

Interpretation: The flow variables measure why benchmarks matter. If investors reward performance relative to self-declared benchmarks, managers have incentives to choose benchmarks that make performance look stronger.

3 Models and empirical specifications

3.1 Time trend in benchmark mismatch

The main trend regression is

$$\text{Mismatch}_{i,t} = \alpha_i + \beta \text{Trend}_t + \gamma \text{StdISBreadth}_{i,t} + \delta (\text{Trend}_t \times \text{StdISBreadth}_{i,t}) + u_{i,t}. \quad (1)$$

- α_i are fund fixed effects.
- Trend_t is time since the start of the sample, measured in years.
- β captures the decline in mismatch for average-breadth funds.
- δ captures whether the decline differs for broad funds.

Interpretation: A negative β means mismatch declines over time. A positive δ means the decline is weaker for broad funds, consistent with mismatches being harder to classify as mistakes for broad strategies.

3.2 Bias and variance specification

For funds with mismatched benchmarks, the paper estimates regressions of the form

$$Y_{i,t} = \alpha + \beta \text{StdISBreadth}_{i,t} + \Gamma' X_{i,t} + \lambda_{c,t} + \varepsilon_{i,t}, \quad (2)$$

where $Y_{i,t}$ is either Bias or Variance, $X_{i,t}$ includes controls, and $\lambda_{c,t}$ are Morningstar-category-by-time fixed effects.

Interpretation: If specialized funds are gaming, low IS Breadth should be associated with higher bias. If broad funds use mismatched benchmarks for hedging or category ambiguity, high IS Breadth should be associated with lower bias and possibly higher variance.

3.3 Benchmark factor-loading and return differences

For funds that switch benchmarks, the paper compares the initial and final benchmark:

$$\Delta \text{BenchmarkCharacteristic}_i = \text{InitialBenchmark}_i - \text{FinalBenchmark}_i. \quad (3)$$

The characteristics include market beta, SMB beta, HML beta, and 36-month return difference.

Interpretation: If mismatched specialized funds initially choose easier benchmarks, their initial benchmarks should have lower market or size exposure or lower subsequent returns than the final matched benchmark.

3.4 Investor-flow regressions around benchmark switches

The paper estimates flow regressions around benchmark switches, separating specialized and broad funds:

$$\begin{aligned} \text{Flow}_{i,t+1} = & \alpha + \beta_1(\text{Raw} - \text{OldBMReturn})_{i,t} + \beta_2(\text{Raw} - \text{NewBMReturn})_{i,t} \\ & + \beta_3\text{PostSwitch}_{i,t} + \beta_4\text{CorrectNewBM}_{i,t} + \beta_5\text{MismatchedOldBM}_{i,t} + \Gamma'X_{i,t} + \varepsilon_{i,t}. \end{aligned} \quad (4)$$

Interpretation: The key question is whether investors continue to use the old benchmark or learn to respond to the new one. For specialized funds, flows become more sensitive to the new matched benchmark after correction.

3.5 Investor learning and benchmark correction

The investor-learning test estimates probit or ordered-probit models:

$$\Pr(\text{ChangeToMatched}_{i,t+1} = 1) = \Phi(\alpha + \beta \text{FlowBetaDifference}_{i,t} + \gamma \text{StdISBreadth}_{i,t} + \delta \text{FlowBetaDifference}_{i,t} \times \text{StdISBreadth}_{i,t}). \quad (5)$$

Interpretation: A positive β means funds are more likely to correct benchmarks when investor flow sensitivity to the correct benchmark rises relative to the self-declared benchmark. A negative interaction means this learning channel is stronger for specialized funds.

3.6 Governance by institutional investors

The paper studies institutional twins using regressions such as

$$\Pr(\text{Mismatch}_{i,t} = 1) = \Phi(\alpha + \beta \text{HasTwin}_{i,t} + \delta \text{HasTwin}_{i,t} \times \text{ISBreadth}_{i,t} + \Gamma'X_{i,t}). \quad (6)$$

Interpretation: Institutional investors are more sophisticated monitors. If governance works, retail funds with institutional twins should be less likely to use mismatched benchmarks, especially when their strategy is specialized.

3.7 Competition and product-positioning tests

The competition regressions relate benchmark switches to benchmark-category competition:

$$\Pr(\text{Switch}_{i,t+1} = 1) = \Phi(\alpha + \beta \text{Competition}_{b,t} + \delta \text{Competition}_{b,t} \times \text{StdISBreadth}_{i,t} + \Gamma' X_{i,t}). \quad (7)$$

Interpretation: Benchmark changes are also product-market decisions. A fund facing many index-fund competitors in its current benchmark category may switch to differentiate itself and to make its risk position more transparent.

3.8 Relative return, tracking error, and switching

The paper estimates probit models in which the dependent variable is whether a fund changes benchmark, changes to a matched benchmark, or changes to a mismatched benchmark. The key explanatory variables are

$$\text{BMAAdjRetDiff(New-Old)} \quad \text{and} \quad \text{TrackErrDiff(Correct-Old)}.$$

Interpretation: A positive coefficient on relative-performance improvement means funds are attracted to benchmarks that make past performance look better. A negative coefficient on tracking-error difference means funds are attracted to benchmarks that better match their actual risk.

4 Identification and empirical design

4.1 What identifies the central facts

- The time-series decline in mismatch comes from comparing benchmark status across repeated Morningstar snapshots.
- The intensive-margin correction comes from tracking funds that existed in 2008 and observing whether they later switch to matched benchmarks.
- IS Breadth allows the paper to split cases where mismatch likely reflects opportunistic benchmark choice from cases where matching is inherently ambiguous.

Interpretation: The paper does not rely on a single causal shock. Its contribution is a carefully documented dynamic market equilibrium: benchmark distortions decline as investors, institutions, and competitors discipline benchmark choice.

4.2 Threats and how the paper addresses them

- **Classification error.** Morningstar style boxes may not perfectly capture a fund’s true strategy. IS Breadth directly addresses this by identifying funds whose strategies span multiple boxes.
- **Changing fund styles.** A benchmark switch could simply follow a style change. Table 2 shows many switches do not coincide with Morningstar style changes, and the paper controls for category-by-time effects.
- **Flow-performance endogeneity.** Flow sensitivity may be shaped by benchmark choice. The paper studies rolling flow betas and switching events to assess how investor learning evolves.
- **Institutional twin selection.** Funds with institutional twins may differ from funds without twins. The paper uses twin-pair-by-time fixed effects when comparing institutional and retail versions.

Interpretation: The design is best read as evidence on mechanisms in a changing market rather than as a single randomized causal experiment. The strength comes from the alignment across trends, switching behavior, flow response, institutional monitoring, competition, and managerial entrenchment.

5 Tables and figures

5.1 Table 1 and Table 2: sample, benchmark concentration, and switch types

Read:

- Table 1 reports the timing of Morningstar benchmark snapshots and summary statistics for the monthly fund panel.
- The full sample has roughly 268,918 fund-month observations. The average benchmark mismatch rate is about 32.6%.
- Average IS Breadth is 0.632, meaning that a large part of active fund holdings can lie outside a single style box.
- Table 2 shows the dominance of the S&P 500 in 2008 and its decline by 2020: the S&P 500's benchmarked asset share falls from 52.4% to 38.7%.
- Benchmark switches are not rare: the authors observe 806 benchmark changes. Of these, 309 are mismatched-to-correct, 139 are mismatched-to-mismatched, 136 are correct-to-mismatched, and 222 are correct-to-correct.

Economic message: The benchmark market becomes less concentrated and more style-specific over time. The table also makes clear that the main story is not merely new funds choosing better benchmarks; many existing funds actively correct prior benchmark choices.

Interpretation: Table 1 tells us that mismatch is a common equilibrium object, not an isolated compliance error. Table 2 adds two important layers. First, benchmark choice is part of product identity: the S&P 500 was historically a convenient broad-market benchmark, but funds migrate toward more specialized indices as investor sophistication and benchmark supply improve. Second, the fact that mismatched-to-correct switches are the single largest switch category supports the paper's claim that market forces can repair benchmark distortions.

Table 1

Summary Statistics. Panel A describes the timing of the snapshots of benchmarks of active equity mutual funds in our sample. The benchmarks reported at these snapshots are then matched to the funds' corresponding prospectus reporting dates for the timing of their exact release. Panel B describes the summary statistics of fund-month observations of these equity funds. Our sample consists of equity mutual funds between December 2008 and December 2020. *Benchmark Mismatch* indicates whether the mutual fund's self-designated benchmark has a category that is different from its Morningstar designated holding category. *Benchmark Change* indicates a change in the self-designated fund benchmark from 12 months prior. *IS Breadth* measures the breadth of a fund's investments and is defined as the percent of a fund's equity holdings that is outside of its principal strategy (style box) as identified by Morningstar. *Standardized IS Breadth* is *IS Breadth* normalized by its mean and standard deviation. *Bias* is the average difference in 36-month returns between the fund's best-fitting benchmark (among the set of 12 most popular equity benchmarks) and the fund's currently selected benchmark. *Variance* is as the squared difference between the monthly returns of the self-declared benchmark and those of the best-fitting benchmark, calculated over a 36-month period. *Large*, *Small*, *Growth*, and *Value* are indicators of the size and growth-value orientation as indicated by the Morningstar equity style box. *Size* is the dollar assets under management by the mutual fund. *Family Size* is the dollar assets under management in the equity fund sector by the family of the mutual fund. *Age* is calculated as the number of years since fund inception. *Fund Load Share* is the fraction of fund family assets sold with front or back load. *Percent Flow* (%) is the monthly net dollar flow into the mutual fund divided by lagged fund size. *Change in Market Share* (%) is the one-month percentage change in the market share of a mutual fund in total the equity fund sector. *Benchmark Adj Return 36m* (%) is the 36-month cumulative net fund return subtracted by the 36-month cumulative total benchmark return. *FF Alpha 36m* (%) is the 36-month cumulative *Fama-French* (1993) adjusted fund return where factor exposures are estimated using 36-month rolling windows. *Star Rating* is the monthly Morningstar Overall Rating. *Flow Beta Difference* is the incremental sensitivity of a fund's rolling 36-month beta regressing next 12-month *Percent Flow* on past 36-month returns adjusted by a matched benchmark over the beta obtained using its self-designated benchmark. *Net Expense Ratio* (%) is the average net expense ratio of the equity fund.

Panel A.

Snapshot Date	Snapshot Date			
December 31, 2008	November 24, 2015			
December 31, 2009	October 13, 2017			
March 5, 2010	August 27, 2018			
September 17, 2012	September 4, 2019			
September 11, 2013	December 7, 2020			
June 5, 2014				

Panel B.

	Full Sample			
	N	Mean	Median	SD
Benchmark Mismatch	268,918	0.326	0	0.469
Benchmark Change	266,422	0.0337	0	0.181
IS Breadth	260,208	0.632	0.652	0.163
Standardized IS Breadth	260,208	0	0.10	1
Bias (%)	267,799	0.022	0	0.134
Variance (%) ²	267,799	0.663	0.022	1.202
Large	268,918	0.544	1	0.498
Small	268,918	0.248	0	0.432
Growth	268,918	0.410	0	0.492
Value	268,918	0.258	0	0.438
Size (\$ million)	268,918	2,019	325	11,920
Family Size (\$ million)	268,918	69,290	11,460	208,300
Age (year)	264,540	17.60	15	13.08
Fund Load Share	268,908	0.372	0.118	0.405
Percent Flow (%)	267,662	-0.379 %	-0.571 %	4.82 %
Change in Market Share (%)	267,855	-0.205 %	-0.379 %	5.32 %
Benchmark Adj Return 36m (%)	267,808	-3.22 %	-3.19 %	13.30 %
FF Alpha 36m (%)	268,865	-8.20 %	-8.21 %	10.7 %
Star Rating	268,918	2.307	2.880	1.570
Rolling 36-Month Flow Beta	258,504	0.0607	0.0132	0.449
Flow Beta Difference	244,028	0.118	0	0.511
Net Expense Ratio (%)	268,282	1.190 %	1.190 %	0.565 %

Table 1: sample and variables.**Table 2**

Benchmark Summary Statistics. Panel A. The largest ten benchmarks by assets at the beginning (December 2008) and the end (December 2020) of our sample. This panel lists the largest indices by dollars of benchmarked assets for the cross-section of mutual funds on December 2008 (top) and December 2020 (bottom). Panel B. Fund Month Observations Mismatched by Size and Style. This panel tabulates the observations by their portfolio composition in the column and benchmark types in the row as related to size (top) and growth-value orientations (bottom). Panel C. Benchmark switches summary. The 806 benchmark changes observed in the sample are listed with the accompanying changes to their *Mismatch* status. 95 of these benchmark changes are accompanied with changes in Morningstar portfolio style box.

Panel A. Largest Benchmarks by Assets

Rank	Benchmark Name	Assets Dec 2008 (\$ Billion)	Fraction of Total Assets Dec 2008	No. Funds
1	S&P 500	960.0	52.4 %	759
2	Russell 1000 Value	161.6	8.8 %	187
3	Russell 2000	122.0	6.7 %	214
4	Russell 1000 Growth	118.3	6.5 %	197
5	Russell Mid Cap Value	51.9	2.8 %	73
6	Russell 3000 Growth	48.3	2.6 %	40
7	Russell 2000 Value	45.3	2.5 %	118
8	Russell Mid Cap Growth	45.1	2.5 %	115
9	Russell 3000	39.9	2.2 %	60
10	S&P Mid Cap 400	36.5	2.0 %	48

Panel B. Mismatches by Size and Style Targets

	Large	Mid	Small				
Benchmark Size Target	Large/Broad	167,752	99.57 %	15,871	24.54 %	3,083	3.97 %
	Mid	509	0.30 %	46,044	71.18 %	2,959	3.81 %
	Small	218	0.13 %	2,772	4.29 %	71,633	92.22 %
	Total	168,479		64,687		77,675	

Growth-Value Orientation Indicated by Holdings

	Blend	Growth	Value				
Benchmark Growth-Value Orientation	Broad/Blend	90,831	85.74 %	51,430	41.43 %	25,589	31.68 %
	Growth	634	0.60 %	72,008	58.01 %	105	0.13 %
	Value	14,468	13.66 %	686	0.55 %	55,090	68.19 %
	Total	105,933		124,124		80,784	

Panel C.

	Observations	% of Total	# With MS Style Changes
Mismatched to Correct	309	38.34 %	20
Mismatched to Mismatched	139	17.25 %	20
Correct to Mismatched	136	16.87 %	9
Correct to Correct	222	27.54 %	46
Total	806	100 %	95

Table 2: benchmark shares and switch types.

5.2 Table 3: validation of IS Breadth

Read:

- Panel A summarizes alternate breadth measures: broad fund name, style drift, broad mandate, and mandate with allocation.
- Panel B shows that standardized IS Breadth is positively related to broad names, style drift, and broad mandates.
- IS Breadth is negatively related to mandates with explicit allocation percentages, consistent with those funds being more specific rather than more open-ended.

Economic message: IS Breadth is not just a mechanical holdings statistic. It correlates with independent signs that a fund's strategy is hard to place in a single style box.

Interpretation: The table is important because the paper’s interpretation depends on separating broad from specialized strategies. If IS Breadth were noisy or unrelated to mandate complexity, the later heterogeneity results would be hard to interpret. Instead, Table 3 shows that the measure lines up with name-based, mandate-based, and holdings-based indicators of breadth.

Table 3
Validation of *IS Breadth* Using Other Measures of Portfolio Breadth. Panel A presents summary statistics of other measures of portfolio breadth. Panel B shows OLS regressions of other measures of portfolio breadth on *Standardized IS Breadth*. Observations are at the fund-by-month level. *Std IS Breadth* is the fraction of a fund’s portfolio that is invested in style boxes outside of the fund’s Morningstar style box, normalized by mean and standard deviation. *Name Broad* is an indicator that equals one if the fund name does not contain “Growth”, “Value”, “Large”, “Small”, or “Mid”. *Style Drift* is an indicator that equals one if a fund has a change in its Morningstar style box classification between the current month and one year later. *Mandate Broad* equals one if the investment description of the fund contains both “Growth” and “Value” or if it contains both “Large” and “Small”. *Mandate w/ Alloc.* indicates if the mandate contains a percentage sign to indicate asset allocation goals. Control variables, whose coefficients are omitted, include the natural logarithms of fund size, fund family size, fund age measured in years, and the fraction of the fund family assets that are sold with loads. All regressions include time-by-portfolio category fixed effects. The standard errors, reported in parenthesis, are clustered at the fund level. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

Panel A.						
	N			Mean		
Name Broad	258,124			0.333		
Style Drift	231,021			0.047		
Mandate Broad	258,124			0.284		
Mandate w/ Alloc.	258,124			0.763		
Panel B.						
	(1)	(2)	(3)	(4)	(5)	(6)
	Name Broad		Style Drift		Mandate	Mandate
					Broad	w/ Alloc.
Std IS Breadth	0.084*** (0.009)	0.007*** (0.003)	0.051*** (0.002)	0.059*** (0.003)	0.041*** (0.009)	-0.045*** (0.009)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
MS Box + Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Fund FE	No	Yes	No	Yes	No	No
Observations	258,124	258,113	224,756	224,730	258,124	258,124
R-squared	0.155	0.931	0.060	0.199	0.032	0.057

our novel panel data allows us to track the systematic changes in 4.1.2. Estimation of linear trends

Table 3: IS Breadth validation.

5.3 Figures 1–3 and Table 4: benchmark mismatch declines over time

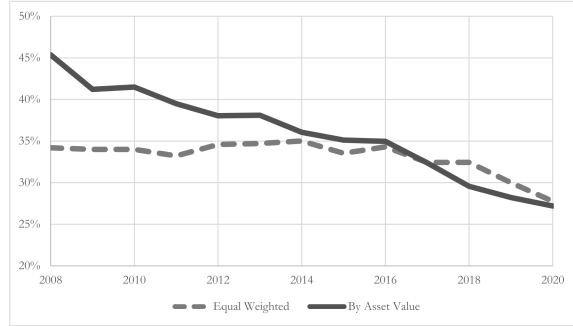
Read:

- Figure 1 shows a steady decline in mismatched benchmarks. Asset-weighted mismatch falls from about 45% in 2008 to about 28% in 2020.
- The intensive-margin panel shows a similar decline among funds that already existed in 2008, indicating correction by existing funds.
- Figure 2 shows the decline in S&P 500 benchmarking, both equal-weighted and asset-weighted.
- Figure 3 shows that about 58% of 2008 mismatched funds that survive to 2020 correct their benchmarks.
- Table 4 estimates a mismatch decline of roughly 0.762 percentage points per year, or 0.771 percentage points with IS Breadth interactions.
- The positive interaction between trend and standardized IS Breadth means broad funds experience a slower decline in mismatch.

Economic message: The market is moving toward more accurate and more differentiated relative performance evaluation. But the correction is concentrated among specialized funds for which a matched benchmark is easier to identify.

Interpretation: These visuals are the paper’s core time-series evidence. Figure 1 rules out the idea that benchmark mismatch is a static or permanent feature of mutual funds. Figure 2 links the correction to a shift away from the default broad-market benchmark. Figure 3 shows that many original offenders correct, which points to learning and competition rather than only sample turnover. Table 4 then formalizes this visual pattern and shows that strategy breadth moderates the speed of correction.

Panel A. Fraction of Active Funds with Mismatched Benchmarks



Panel B. Intensive Margin – Fraction of Active Funds from 2008 with Mismatched Benchmarks

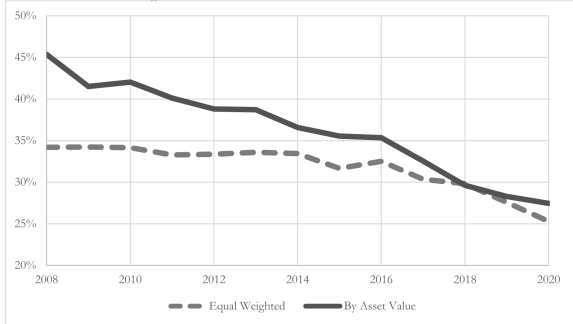


Figure 1: mismatch trends and intensive margin.

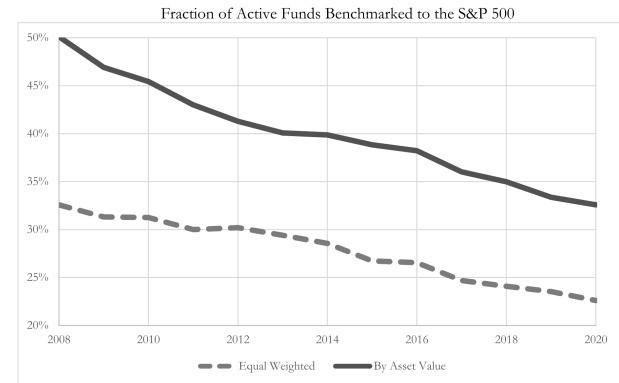


Figure 2: Fraction of Active Funds Benchmarked to the S&P 500. This figure shows the equal weighted and the asset value weighted measures of active funds benchmarked to the S&P 500 over time.

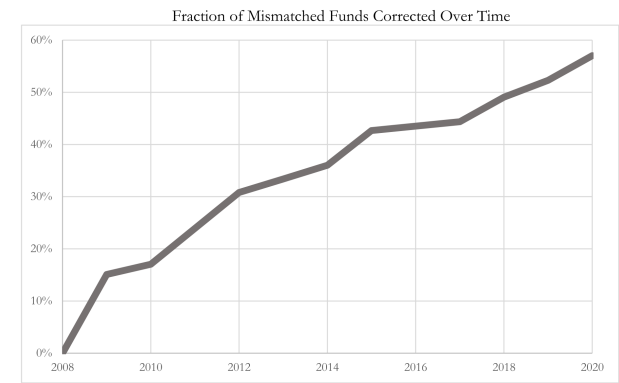


Figure 3: Fraction of Mismatched Funds Corrected Over Time. This figure shows the fraction of mismatched funds corrected over time from December 2008. This figure plots the fraction of funds with matched benchmarks over time.

Figures 2–3: S&P 500 decline and corrections.

Table 4

Time Trend in Benchmark Mismatch. This table regresses the indicator for *Benchmark Mismatch* against *Trend*, *Standardized IS Breadth*, and their interaction. Observations are at the fund-by-month level. *Benchmark Mismatch* indicates whether the mutual fund's self-designated benchmark has a style that is different from its Morningstar designated holding style. *Trend* is the total time since the start of the sample, calculated as the number of years and the fractional equivalent of months (each month accounts for as 1/12th of a year). *Std IS Breadth* is the breadth of a fund's investments and is defined as the percent of the fund's equity holdings that are outside of its portfolio style box as identified by Morningstar, normalized by mean and standard deviation. *Trend X Std IS Breadth* is the interaction of the prior two variables. The standard errors, reported in parenthesis, are clustered at the fund level. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

	(1) Benchmark Mismatch	(2)	(3)
Trend (Year + Month/12)	–0.00762*** (0.00124)	–0.00778*** (0.00125)	–0.00771*** (0.00126)
Std IS Breadth		–0.00585 (0.00459)	–0.0237*** (0.00102)
Trend X Std IS Breadth			0.00289*** (0.000930)
Fund FE	Yes	Yes	Yes
Observations	268,896	260,188	260,188
Adjusted R-squared	0.723	0.723	0.723

Table 4: linear trend in mismatch.

5.4 Table 5: investor flows respond to benchmark-adjusted returns

Read:

- Raw minus benchmark return is positively related to all three flow measures: change in market share, positive flow, and percent flow.
- The relation remains after controlling for 36-month alpha and Morningstar star ratings.
- Star ratings also matter, but the self-declared benchmark-adjusted return has independent explanatory power.

Economic message: Investors care about performance relative to the benchmark displayed to them. Therefore, benchmark choice affects perceived performance and fund flows.

Interpretation: Table 5 provides the incentive foundation for the rest of the paper. If investors did not respond to self-declared benchmark-adjusted performance, benchmark mismatch would be less economically meaningful. Instead, the table shows that benchmark choice shapes flows even after controlling for standard performance metrics. This creates both a moral-hazard problem and a reason for market participants to discipline benchmark choice.

Determinants of Monthly Fund Flow. This table reports regressions of measures of investor flow on past 36-month benchmark-adjusted performance. % Change in Market Share is the percent change in the share of the fund over the total equity mutual fund market. Positive Flow indicates that net dollar flow is positive. % Flow is the net dollar flow divided by lagged fund assets, winsorized at the 1 % level at each tail per cross-section. The dependent variables are monthly measures in month $t + 1$. Raw - Benchmark Return is the difference in the cumulative 36-month net return between the fund and its self-designated benchmark. 36 Month Alpha (%) is the 36-month cumulative Fama-French (1993) adjusted fund return where factor exposures are estimated using 36-month rolling windows. Star Rating is Overall Fund Rating from Morningstar. Other variables are defined in Appendix A. All independent variables are measured in (up to) month t . The standard errors, reported in parenthesis, are clustered at the fund level. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

	(1)	(2)	(3)	(4)	(5)	(6)
	% Change in Market Share	% Change in Market Share	Positive Flow	Positive Flow	% Flow	% Flow
Raw - Benchmark Return	0.346*** (0.0432)	0.430*** (0.0601)	0.0363*** (0.00342)	0.0383*** (0.00390)	0.0383*** (0.00395)	0.0415*** (0.00465)
36 Month Alpha	0.899*** (0.0534)	0.674*** (0.0609)	0.0540*** (0.00364)	0.0490*** (0.00392)	0.0582*** (0.00441)	0.0526*** (0.00476)
Star Rating	0.0575*** (0.00405)	0.0708*** (0.00396)	0.00288*** (0.000249)	0.00353*** (0.000255)	0.00331*** (0.000310)	0.00440*** (0.000312)
Log Fund Size		-0.00393* (0.00238)		-0.000832*** (0.000134)		-0.00148*** (0.000189)
Log Family Size		-0.00666*** (0.00158)		0.000215** (9.22e-05)		0.000318*** (0.000121)
Expense Ratio		0.00578 (0.00835)		0.000899* (0.000461)		0.00223*** (0.000789)
Log Age		-0.108*** (0.00549)		-0.00273*** (0.000296)		-0.00278*** (0.000345)
Load Fund Share		-0.0153* (0.00906)		-0.000754 (0.000498)		-0.00127* (0.000667)
MS Category X Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	266,505	261,731	266,695	261,918	266,505	261,731
Adjusted R-squared	0.125	0.157	0.185	0.190	0.066	0.073

Table 5: investor flows and benchmark-adjusted performance.

5.5 Table 6 and Table 7: gaming versus broad-strategy interpretation

Read:

- Table 6 shows that higher IS Breadth is associated with lower benchmark bias among mismatched funds.
- This means specialized mismatched funds tend to have more favorable benchmark bias, while broad funds have less bias.
- Table 7 shows that mismatched funds initially select benchmarks with lower market beta and lower SMB beta than the final benchmark they later adopt.
- The return differences in Table 7 suggest that initial benchmarks tend to underperform the final benchmarks, making the fund look better relative to the initial benchmark.

Economic message: Mismatch has different meanings. For specialized funds, it can look like strategic selection of an easier benchmark. For broad funds, mismatch appears less associated with performance bias and may reflect genuine difficulty in finding a single benchmark.

Interpretation: Table 6 is the main evidence against treating all mismatch as cheating. Broad funds do not display the same favorable bias. Table 7, however, strengthens the cheating interpretation for specialized funds: the initial benchmark is not only stylistically mismatched but also appears to have lower factor exposures and lower subsequent returns. That is exactly the pattern a manager would want if the objective were to look better in relative performance terms.

Table 6

Bias, Variance, and IS Breadth among Mismatched Funds. This table reports regressions of statistical measures of *Bias* and *Variance* on *Standardized IS Breadth* and various controls. Observations are at the fund-by-month level but composed of only funds with mismatched benchmarks. *Std IS Breadth* is the breadth of a fund's investments and is defined as the percent of the fund's equity holdings that are outside of its portfolio style box as identified by Morningstar, normalized by mean and standard deviation. We define *Bias* as the average difference in 36-month returns between the fund's best-fitting benchmark (among the set of 12 most popular equity benchmarks) and the fund's currently selected benchmark. *Variance* is as the squared difference between the monthly returns of the self-declared benchmark and those of the best-fitting benchmark, calculated over a 36-month period. The standard errors, reported in parenthesis, are clustered at the fund level. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

	(1) Bias (%)	(2)	(3) Variance (% ²)	(4)
Std IS Breadth	-0.0132*** (0.00232)	-0.0127*** (0.00228)	-0.0217 (0.0260)	-0.0197 (0.0264)
Log Fund Size		0.00210 (0.00144)		0.0436** (0.0178)
Log Family Size		0.00245** (0.000965)		-0.0136 (0.0121)
Expense Ratio		0.00820 (0.00499)		0.202*** (0.0705)
Log Age		-0.00202 (0.00326)		-0.0463 (0.0356)
Load Fund Share		-0.0103** (0.00523)		-0.0620 (0.0698)
MS Category X Time FE	Yes	Yes	Yes	Yes
Benchmark FE	Yes	Yes	Yes	Yes
Observations	84,669	83,807	84,669	83,807
Adjusted R-squared	0.434	0.437	0.359	0.362

Table 6: bias and variance by IS Breadth.

Table 7

Difference in Benchmark Factor Loadings and Returns between Initial and Final Benchmark. This table uses the sample of funds that changed benchmarks. The dependent variable in Panel A is the difference in the factor loadings between the initial benchmark and the final benchmark in the 36 months post inception. *Mismatch* indicates whether the mutual fund's self-designated benchmark has a category that is different from its Morningstar designated holding category. *Std IS Breadth* is the breadth of a fund's investments and is defined as the percent of the fund's equity holdings that are outside of its portfolio style box as identified by Morningstar, normalized by mean and standard deviation. *Market Beta Difference* is the difference between the betas of initial benchmark and the final benchmark returns against the Market Risk Factor in the FF 3-factor regression. *SMB Beta Difference* is the difference between the betas of initial benchmark and the final benchmark returns against the SMB Risk Factor in the FF 3-factor regression. *HML Beta Difference* is the difference between the betas of initial benchmark and the final benchmark returns against the HML Risk Factor in the FF 3-factor regression. The dependent variable in Panel B is the difference in the raw returns between the initial benchmark and the final benchmark in the 36 months post inception. *Return Difference* is the 36 month return of the initial benchmark minus that of the final benchmark. *Return Diff Systematic* is the systematic portion of the 36-month Return Difference measured as the portion of the return that could be explained by a FF 3-factor model. *Return Diff Non-Systematic* is the non-systematic/residual portion of the 36-month Return Diff not explained by the FF 3-factor model. The standard errors, reported in parenthesis, are double clustered at the fund and date level. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

Panel A. Difference in Benchmark Systematic Risk Loadings Post Inception (Initial – Final Benchmark)					
	(1) Market Beta Difference	(2)	(3) SMB Beta Difference	(4)	(5) HML Beta Difference
Mismatch	-0.0387*** (0.00803)	-0.0424*** (0.00596)	-0.131*** (0.0175)	-0.145*** (0.0172)	0.0523** (0.0245)
Mismatch *		0.000220		0.0723*** (0.0148)	-0.0476*** (0.0216)
Std. IS Breadth		(0.000837)		-0.0391*** (0.0141)	(0.00825)
Std. IS Breadth		0.000184 (0.000499)		-0.0391*** (0.0141)	0.0158*** (0.00362)
Observations	794	659	794	659	659
Adjusted R-squared	0.057	0.063	0.033	0.044	0.010
Panel B. Difference in 36-Month Benchmark Return Post Inception (Initial – Final Benchmark)					
	(1) Return Difference	(2)	(3) Return Diff Systematic	(4)	(5) Return Diff Non-Systematic
Mismatch	-0.0443*** (0.0150)	-0.0476*** (0.0138)	-0.0452*** (0.0129)	-0.0510*** (0.0125)	0.000816 (0.00255)
Mismatch *		0.0123***		0.0233***	-0.0110***
Std. IS Breadth		(0.00169)		(0.00333)	(0.00242)
Std. IS Breadth		-0.00148 (0.00222)		-0.00721*** (0.00159)	0.00573*** (0.00220)
Observations	794	659	794	659	659

Table 7: initial minus final benchmark exposures.

5.6 Table 8 and Figure 4: investor flow sensitivity before and after benchmark switches

Read:

- Table 8 studies monthly observations around benchmark switches from 24 months before to 24 months after the switch.
- For specialized funds, flows respond to both old and new benchmark-adjusted returns, but after a correction investors become more sensitive to the new matched benchmark.
- Broad funds display weaker evidence that benchmark corrections change investor response, consistent with investors being less certain about the appropriate benchmark for broad strategies.
- Figure 4 shows that rolling flow sensitivity to benchmark-adjusted returns is much larger for correctly benchmarked funds than for mismatched funds.
- By the end of the sample, the median flow beta for correctly benchmarked funds rises sharply, while the mismatched-fund flow beta remains much lower.

Economic message: Investor learning helps correct benchmark mismatch. The more investors reward performance relative to the correct benchmark, the larger the fund's incentive to adopt it.

Interpretation: Table 8 and Figure 4 are especially important because they connect benchmark choice to investor behavior. They show not only that funds change benchmarks, but also that investors process those changes. The difference between specialized and broad funds again matters: learning disciplines cases where the correct benchmark is clear, but its force is weaker when the fund's strategy is legitimately hard to benchmark.

Table 9

Investor Learning as a Determinant for Benchmark Correction. This table estimates probit and ordered probit models of the determinants for benchmark corrections. *Change to Correct* indicates a benchmark switch in the following 12 months and the new benchmark is matched to the fund portfolio style box. In the *Ordered Probit* model, the left-hand side variable is 1 if fund changes to a correctly matched benchmark, 0 if there is no change in the benchmark, and -1 if the fund changes to a mismatched benchmark. *Flow Beta Difference* is the incremental sensitivity of a fund's rolling 36-month flow beta on the past 36-month returns over a matched benchmark, compared with the flow beta over its self-designated benchmark. *Std IS Breadth* is the breadth of a fund's investments and is defined as the percent of the fund's equity holdings that are outside of its portfolio style box as identified by Morningstar, normalized by mean and standard deviation. The standard errors, reported in parenthesis, are clustered at the fund level. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

	(1) Change to Matched	(2)	(3) Ordered Probit	(4)
Flow Beta Difference	0.0564*** (0.0212)	0.0550** (0.0214)	0.0440* (0.0250)	0.0416* (0.0249)
Std IS Breadth	-0.0549** (0.0258)	-0.0574** (0.0266)	-0.0853*** (0.0190)	-0.0895*** (0.0195)
Std IS Breadth X Flow Beta Difference	-0.0569*** (0.0167)	-0.0582*** (0.0168)	-0.0354* (0.0211)	-0.0347* (0.0208)
36 Month Alpha	-0.645*** (0.226)	-0.582** (0.241)	0.0747 (0.189)	-0.0101 (0.194)
Benchmark Mismatch	0.533*** (0.0472)	0.566*** (0.0487)	0.272*** (0.0406)	0.308*** (0.0413)
Log Fund Size		-0.0520*** (0.0181)		-0.0140 (0.0125)
Log Family Size		0.0386*** (0.0108)		0.0325*** (0.00802)
Expense Ratio		-0.0361 (0.0671)		-0.0382 (0.0239)
Log Age		0.0563 (0.0371)		0.000568 (0.0270)
Load Fund Share		0.172*** (0.0575)		0.0821** (0.0406)
MS Category X Time FE	Yes	Yes	Yes	Yes
Observations	203,929	201,334	233,100	230,384
Pseudo R-squared	0.0854	0.0923	0.0366	0.0412

Table 9: investor learning and correction.**Table 10**

Governance by Institutional Investors and the Likelihood of Mismatched Benchmarks. Observations are at the fund-by-month level. Panel A restricts to twin pairs of retail funds and institutional funds (funds with only institutional share classes) and estimates the differential between institutional and non-institutional fund in the twin pair in *Bias* and *Variance*. Regressions in Panel A follow an OLS model and include twin-pair-by-time fixed effects. Panel B uses the full sample and regresses the incidence of *Benchmark Mismatch* on the existence of institutional twins using a probit model. *Benchmark Mismatch* equals one if a fund's self-designated benchmark has a classification that is different from its Morningstar designated holding classification. *Has Twin* indicates that a fund has a separate account or institutional fund twin. *Std IS Breadth* is the breadth of a fund's investments and is defined as the percent of the fund's equity holdings that are outside of its portfolio style box as identified by Morningstar, normalized by mean and standard deviation. Control variables, whose coefficients are omitted, include the natural logarithms of fund size, fund family size, fund age measured in years, and the fraction of the fund family assets that are sold with loads. Regressions in Panel B include time-by-portfolio category fixed effects. Column 2 further includes fund fixed effects. Standard errors are clustered by twin pair in Panel A and by fund in Panel B and are reported in parentheses. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

Panel A.		
	(1) Bias	(2) Variance
Institutional	-0.006** (0.002)	-0.078** (0.031)
Twin Pair X Time FE	Yes	Yes
Observations	123,700	123,700
R-squared	0.694	0.641
Panel B.		
	(1) Benchmark Mismatch	(2)
Has Twin	-0.129*** (0.049)	-0.061 (0.232)
Has Twin X IS Breadth	0.149*** (0.043)	0.209** (0.083)
Std IS Breadth	0.308*** (0.034)	0.129** (0.059)
Control Variables	Yes	Yes
MS Category X Time FE	Yes	Yes
Fund FE	No	Yes
Observations	257,952	95,618
Pseudo R-squared	0.157	0.521

Table 10: institutional monitoring.

5.8 Tables 11–13: competition, product characteristics, and managerial entrenchment

Read:

- Table 11 shows that higher index-fund competition in a benchmark category increases benchmark switching and switching to matched benchmarks.
- The effect is weaker for high-IS-Breadth funds, again suggesting that broad funds face more benchmark ambiguity.
- Table 12 shows that funds are more likely to switch when a new benchmark improves relative performance or reduces tracking error.
- Table 13 shows that manager tenure and the share of family assets managed by the manager are positively associated with mismatch.
- The negative interaction between IS Breadth and managerial power variables suggests that entrenchment matters most for specialized funds, where mismatch is more likely to be opportunistic.

Economic message: Benchmark choice is shaped by product-market forces and managerial incentives. Competition promotes differentiation and correction; entrenchment slows correction.

Interpretation: Table 11 connects benchmark correction to the competitive structure of the benchmark category. Table 12 shows the fund’s objective is multidimensional: a benchmark can improve relative performance presentation and reduce tracking error. Table 13 adds governance frictions: even when market forces push toward correction, entrenched managers may resist, especially when the benchmark mismatch can be useful for performance presentation.

Table 11

Competition at the Benchmark-Category Level and Benchmark Switches. This table reports the estimation of probit models using fund month observations. *Change BM* equals one if a fund switches its benchmark in the following 12-month period, and zero otherwise. *Change to Matched* equals one if a fund switches its benchmark to one that matches with its Morningstar category, and zero otherwise. Measures of competition are at the benchmark-category level and include the natural logarithm of the number of index funds and the market share of index funds. *Std IS Breadth* is the breadth of a fund’s investments and is defined as the percent of the fund’s equity holdings that are outside of its portfolio style box as identified by Morningstar, normalized by mean and standard deviation. *Benchmark Mismatch* equals one if a fund’s self-designated benchmark has a classification that is different from its Morningstar designated holding classification. Control variables, whose coefficients are omitted, include the natural logarithms of fund size, fund family size, fund age measured in years, and the fraction of the fund family assets that are sold with loads. All regressions include time-by-portfolio category fixed effects. The standard errors, reported in parenthesis, are clustered at the benchmark category level. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

	(1) Change BM	(2)	(3) Change to Matched	(4) Change to Matched
ln(# Index Funds)	0.157*** (0.007)		0.167*** (0.008)	
ln(# Index Funds) X Std IS Breadth	-0.050*** (0.006)		-0.050*** (0.007)	
Index Fund Share		0.976*** (0.073)		1.223*** (0.083)
Index Fund Share X Std IS Breadth		-0.348*** (0.063)		-0.313*** (0.073)
Std IS Breadth	0.111*** (0.021)	0.030** (0.014)	0.033 (0.027)	-0.056*** (0.017)
Benchmark Mismatch	0.344*** (0.016)	0.427*** (0.016)	0.481*** (0.019)	0.548*** (0.018)
Controls	Yes	Yes	Yes	Yes
MS Category X Time FE	Yes	Yes	Yes	Yes
Pseudo R-squared	0.0916	0.0858	0.107	0.102
Observations	191,823	191,823	187,922	187,922

funds de-biases investors, index fund competition is associated with investor learning which also leads to benchmark corrections.

Table 11: competition.

Table 12

Relative Performance, Tracking Errors and Benchmark Switches. This table reports the estimation of probit models using fund month observations. *Change BM* (Change to Matched, Change to Mismatched) equals one if a fund switches its benchmark (switches to a benchmark that matches its portfolio, switches to a benchmark that mismatches its portfolio) in the subsequent 12-month period, and zero otherwise. *BM Adjusted Diff* (New-Old) is the difference between the new benchmark adjusted return and the old benchmark adjusted return, calculated as the monthly average during the past 36 months, and equal to zero if the benchmark does not switch. *Tracker Diff* (Correct-Old) is the difference between the tracking error around the holdings-implied benchmark and the tracking error around the old benchmark. Each tracking error is calculated as the standard deviation of monthly return differentials between the fund and a benchmark (correct benchmark or old benchmark, respectively, zero if the old benchmark is correct) over the past 36 months. *Std IS Breadth* is the breadth of a fund’s investments and is defined as the percent of the fund’s equity holdings that are outside of its portfolio style box as identified by Morningstar, normalized by mean and standard deviation. Control variables, whose coefficients are omitted, include the natural logarithms of fund size, fund family size, fund age measured in years, and the fraction of the fund family assets that are sold with loads. All regressions include time-by-portfolio category fixed effects. Standard errors are clustered by fund and reported in parentheses. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

	(1) Change BM	(2) Change to Matched	(3) Change to Mismatched	(4) Change BM	(5) Change to Matched	(6) Change to Mismatched
BM Adjusted Diff (New-Old)	2.107*** (0.422)	0.938* (0.481)	3.374*** (0.705)			
BM Adjusted Diff (New-Old) X Std IS Breadth	0.772** (0.397)	0.751 (0.470)	0.079 (0.691)			
Tracker Diff (Correct-Old)				-0.580*** (0.062)	-0.608** (0.074)	-0.284*** (0.082)
Tracker Diff (Correct-Old) X Std IS Breadth				0.275*** (0.054)	0.307*** (0.058)	0.087 (0.067)
Std IS Breadth	0.045** (0.021)	-0.021 (0.024)	0.131*** (0.033)	0.083*** (0.023)	0.025 (0.055)	0.162*** (0.032)
Control Variables	Yes	Yes	Yes	Yes	Yes	Yes
MS Category X Time FE	Yes	Yes	Yes	Yes	Yes	Yes
Pseudo R2	0.0671	0.0510	0.122	0.0707	0.0716	0.0841
Observations	209,693	205,237	178,923	209,693	205,237	178,923

Table 13

extraction considerations. For specialized funds, however, if principals remain naïve and agents choose benchmarks for the purpose of rent

MANAGERIAL ENTRENCHMENT IN BENCHMARK CHOICES: EVIDENCE FROM FUND MONTH OBSERVATIONS. *Benchmark Mismatch* equals one if a fund’s self-designated benchmark has a classification that is different from its Morningstar designated holding classification. *Std IS Breadth* is the breadth of a fund’s investments and is defined as the percent of the fund’s equity holdings that are outside of its portfolio style box as identified by Morningstar, normalized by mean and standard deviation. *Manager Tenure Length* is the (average) number of years that the fund’s manager(s) had worked at the fund family. *% Family Managed* is the (average) fraction of a fund family’s asset under management in equity funds managed by the fund’s manager(s). Control variables, whose coefficients are omitted, include the natural logarithms of fund size, fund family size, fund age measured in years, and the fraction of the fund family assets that are sold with loads. The standard errors, reported in parenthesis, are clustered at the fund level. *, **, *** indicates statistical significance at the 90 %, 95 %, and 99 % level respectively.

	(1) Benchmark Mismatch	(2)	(3)
Manager Tenure Length	0.028*** (0.00464)		0.0238*** (0.00412)
% Family Managed		0.402*** (0.0945)	0.548*** (0.0962)
Std IS Breadth	0.484*** (0.0423)	0.643*** (0.0343)	0.519*** (0.0469)
Std IS Breadth X Tenure Length		-0.00821** (0.00326)	-0.00699** (0.00343)
Std IS Breadth X % Family Managed		-0.148*** (0.0588)	-0.131** (0.0576)
Control Variables	Yes	Yes	Yes
MS Category X Time FE	Yes	Yes	Yes
Observations	253,799	253,799	253,799
Pseudo R-squared	0.166	0.162	0.170

manager, decreasing the incidence of mismatched benchmarks and the

Table 12: return, risk, and switches.

Table 13: managerial entrenchment.

6 Short summary

- The paper studies the evolution of self-declared mutual fund benchmarks from 2008 to 2020.
- Benchmark mismatch is initially high but declines substantially over time.
- The decline is driven by existing funds correcting benchmarks, not only by fund entry and exit.
- The S&P 500 becomes less dominant as funds move toward more differentiated style benchmarks.
- IS Breadth measures how difficult it is to classify a fund into one style box.
- Specialized mismatched funds show evidence consistent with benchmark gaming.
- Broad mismatched funds show less evidence of performance bias and may face real benchmark ambiguity.
- Investors respond to benchmark-adjusted performance, giving managers incentives over benchmark choice.
- Investor learning increases benchmark correction, especially for specialized funds.
- Institutional twins provide governance that improves benchmark quality.
- Competition in benchmark categories induces switches toward more matched benchmarks.
- Funds also switch when new benchmarks improve relative performance or reduce tracking error.
- Managerial entrenchment slows correction of mismatched benchmarks.
- The main lesson is nuanced: benchmark mismatch can reflect both cheating and legitimate competitive positioning.

7 Conclusion

7.1 Core conclusion

Chen, Evans, and Sun show that benchmark mismatch in active equity mutual funds is real and economically important, but it is not a single phenomenon. Some mismatches are consistent with performance gaming, especially among specialized funds whose holdings clearly map to a style benchmark. Other mismatches appear connected to broad strategies, risk hedging, and product-positioning constraints. Over time, market forces reduce mismatch, especially where the correct benchmark is easier to identify.

7.2 What the paper changes in the benchmark-mismatch debate

The paper shifts the discussion away from a binary “honest versus cheating” framing. Prior evidence documented that many funds declare benchmarks that are easier to beat. This paper agrees with that concern for specialized funds, but adds that benchmark mismatch can also be a symptom of limited benchmark technology. A broad fund may not have a single benchmark that simultaneously describes size exposure, value-growth orientation, style drift, risk hedging, and product differentiation. Therefore, regulation and investor evaluation should distinguish opportunistic mismatch from genuine benchmark ambiguity.

7.3 Implications for investors

For investors, the main practical lesson is to look beyond the prospectus benchmark. A fund’s self-declared benchmark should be checked against its holdings-implied Morningstar category, its tracking error, and its IS Breadth. A specialized fund with a broad benchmark is more suspicious than a genuinely broad fund with a benchmark that only partially summarizes its mandate. Investors should also pay attention to benchmark changes because a switch to a more matched benchmark can alter how past and future performance should be interpreted.

7.4 Implications for fund families and managers

The paper suggests that benchmark choice is part of product strategy. Funds compete not only on fees and returns but also on how they present themselves to investors. A benchmark can make the fund look better, but it can also expose the fund to a crowded competitive category. This creates a tradeoff: managers may want a benchmark that improves relative performance, but they also want a benchmark that reduces tracking error and differentiates the product. In equilibrium, benchmark choice reflects marketing, risk management, and governance pressure.

7.5 Implications for regulation

The SEC requirement that funds disclose a broad-based benchmark improved transparency, but the paper shows that broad-based disclosure can be insufficient. If the rule allows the S&P 500 to serve as a default benchmark even when the fund’s holdings are specialized, investors may be misled. A better regulatory approach may require funds to disclose why the chosen benchmark is appropriate, report a holdings-implied benchmark alongside the self-declared benchmark, or disclose benchmark mismatch and tracking error in a standardized way. At the same time, regulation should not force broad funds into artificially narrow benchmarks that misrepresent their strategy.

7.6 Implications for relative performance evaluation

Relative performance evaluation works only if the comparison group is appropriate. The paper shows that the benchmark itself is a strategic object. If the agent can influence the benchmark, the RPE contract can be distorted. But the paper also shows that market participants can improve the benchmark over time. Investor learning, institutional monitoring, and product-market competition act like decentralized governance mechanisms that gradually improve RPE quality.

7.7 Implications for asset-pricing and fund-flow research

Benchmarks affect flows, and flows affect asset demand. Therefore, benchmark selection can matter for prices, especially when many funds cluster around a large index such as the S&P 500. As funds move away from broad benchmarks toward more differentiated benchmarks, old benchmark-induced distortions may fade while new, category-specific demand distortions may emerge. This links benchmark choice to the broader literature on inelastic demand, benchmark-driven trading, and fund flow-performance sensitivity.

7.8 Limitations

The paper’s mismatch measure is intentionally transparent, but no style-box measure can perfectly represent every fund strategy. Some funds may deliberately hold broad portfolios for reasons that are hard to observe. The institutional governance and investor learning tests are persuasive mechanism evidence, but they are not randomized experiments. The paper also focuses on active U.S. equity funds, so the conclusions may differ for fixed-income funds, multi-asset funds, ETFs, or non-U.S. markets.

7.9 Research agenda

Several natural extensions follow. First, future work could study whether benchmark correction changes portfolio holdings or only disclosure. Second, researchers could examine whether benchmark mismatch affects manager compensation, not just investor flows. Third, the IS Breadth idea could be applied to ETFs, target-date funds, ESG funds, and multi-asset funds where benchmark choice is even more ambiguous. Fourth, the interaction between benchmark design and market-wide demand distortions remains important as fund competition becomes more index- and category-based.

7.10 Final takeaway

The paper’s final message is balanced. Benchmark mismatch can be a form of manipulation, but not every mismatch is manipulation. The right economic question is when mismatch creates misleading relative performance evaluation and when it reflects a limited benchmark set for broad investment strategies. Market forces have improved benchmark quality, but they do not eliminate the need for better disclosure, investor sophistication, and careful regulatory design.